

0902.0431: residuals

equations measured 2026-09-05T10:59:49Z against the model of 2026-09-02

Corrected (10)

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
0902.0431_E00425 (was)	70	■ 0.082	$\begin{aligned} & A \times A^* \times Y A^* = A(X \times Y) A^* \\ & \left(PMA^*\right)^*\left(PNA^*\right)=A M^* P^* P N A=A\left(M^* N\right) A, \\ & \left(PMA^*\right)\left(AYA^*\right)=P(MY) A^*, \\ & \overline{PMA^* \times PNA^*}=\overline{{}^t \tilde{P}(M \times N){}^t \tilde{A}^*}}=\overline{P \overline{M} \times \overline{N} A^*}, \text { etc. } \end{aligned}$	$\begin{aligned} & A X A^* \times A Y A^*=A(X \times Y) A^* \\ & \left(P M A^*\right)^*\left(P N A^*\right)=A M^* P^* P N A=A\left(M^* N\right) A, \\ & \left(P M A^*\right)\left(A Y A^*\right)=P(M Y) A^*, \\ & \overline{P M A^* \times P N A^*}=\overline{{}^t \tilde{P}(M \times N){}^t \tilde{A}^*}}=P \overline{M} \times \overline{N} A^*, \text { etc. } \end{aligned}$	$\begin{aligned} & A X A^* \times A Y A^*=A(X \times Y) A^*, \\ & \left(P M A^*\right)^*\left(P N A^*\right)=A M^* P^* P N A=A\left(M^* N\right) A, \\ & \left(P M A^*\right)\left(A Y A^*\right)=P(M Y) A^*, \\ & \overline{P M A^* \times P N A^*}=\overline{{}^t \tilde{P}(M \times N){}^t \tilde{A}^*}}=P \overline{M} \times \overline{N} A^*, \text { etc. } \end{aligned}$
0902.0431_E00425 (now)	70	—	$\begin{aligned} & A \times A^* \times Y A^*=A(X \times Y) A^* \\ & \left(P M A^*\right)^*\left(P N A^*\right)=A M^* P^* P N A=A\left(M^* N\right) A, \\ & \left(P M A^*\right)\left(A Y A^*\right)=P(M Y) A^*, \\ & \overline{P M A^* \times P N A^*}=\overline{{}^t \tilde{P}(M \times N){}^t \tilde{A}^*}}=P \overline{M} \times \overline{N} A^*, \text { etc. } \end{aligned}$	$\begin{aligned} & A X A^* \times A Y A^*=A(X \times Y) A^* \\ & \left(P M A^*\right)^*\left(P N A^*\right)=A M^* P^* P N A=A\left(M^* N\right) A, \\ & \left(P M A^*\right)\left(A Y A^*\right)=P(M Y) A^*, \\ & \overline{P M A^* \times P N A^*}=\overline{{}^t \tilde{P}(M \times N){}^t \tilde{A}^*}}=P \overline{M} \times \overline{N} A^*, \text { etc. } \end{aligned}$	$\begin{aligned} & A X A^* \times A Y A^*=A(X \times Y) A^*, \\ & \left(P M A^*\right)^*\left(P N A^*\right)=A M^* P^* P N A=A\left(M^* N\right) A, \\ & \left(P M A^*\right)\left(A Y A^*\right)=P(M Y) A^*, \\ & \overline{P M A^* \times P N A^*}=\overline{{}^t \tilde{P}(M \times N){}^t \tilde{A}^*}}=P \overline{M} \times \overline{N} A^*, \text { etc. } \end{aligned}$
basis: inferred / ink					
0902.0431_E00425 (was)	70	■ 0.082	$\begin{aligned} & A \times A^* \times Y A^*=A(X \times Y) A^* \\ & \left(P M A^*\right)^*\left(P N A^*\right)=A M^* P^* P N A=A\left(M^* N\right) A, \\ & \left(P M A^*\right)\left(A Y A^*\right)=P(M Y) A^*, \\ & \overline{P M A^* \times P N A^*}=\overline{{}^t \tilde{P}(M \times N){}^t \tilde{A}^*}}=P \overline{M} \times \overline{N} A^*, \text { etc. } \end{aligned}$	$\begin{aligned} & A X A^* \times A Y A^*=A(X \times Y) A^* \\ & \left(P M A^*\right)^*\left(P N A^*\right)=A M^* P^* P N A=A\left(M^* N\right) A, \\ & \left(P M A^*\right)\left(A Y A^*\right)=P(M Y) A^*, \\ & \overline{P M A^* \times P N A^*}=\overline{{}^t \tilde{P}(M \times N){}^t \tilde{A}^*}}=P \overline{M} \times \overline{N} A^*, \text { etc. } \end{aligned}$	$\begin{aligned} & A X A^* \times A Y A^*=A(X \times Y) A^*, \\ & \left(P M A^*\right)^*\left(P N A^*\right)=A M^* P^* P N A=A\left(M^* N\right) A, \\ & \left(P M A^*\right)\left(A Y A^*\right)=P(M Y) A^*, \\ & \overline{P M A^* \times P N A^*}=\overline{{}^t \tilde{P}(M \times N){}^t \tilde{A}^*}}=P \overline{M} \times \overline{N} A^*, \text { etc. } \end{aligned}$
0902.0431_E00425 (now)	70	—	$\begin{aligned} & \&=\nu\Big(\sum_{k=0}^3\lambda_kH_k\Big)\\ & \lambda_kH_k\Big)\&=\frac{1}{2}\big(\lambda_0(-H_0+H_1-H_2+H_3)+\lambda_1(-H_0+H_1+H_2-\\ & \&\quad+\lambda_2(H_0+H_1+H_2+H_3)+\lambda_3(-H_0-H_1+H_2+H_3)\end{aligned}$	$\begin{aligned} & =\nu\left(\sum_{k=0}^3 \lambda_k H_k\right) \\ & =\frac{1}{2}\left(\lambda_0\left(-H_0+H_1-H_2+H_3\right)+\lambda_1\left(-H_0+H_1+H_2-\right.\right. \\ & \quad \left.\left.+\lambda_2\left(H_0+H_1+H_2+H_3\right)+\lambda_3\left(-H_0-H_1+H_2+H_3\right)\right) \end{aligned}$	$\begin{aligned} & A X A^* \times A Y A^*=A(X \times Y) A^*, \\ & \left(P M A^*\right)^*\left(P N A^*\right)=A M^* P^* P N A=A\left(M^* N\right) A, \\ & \left(P M A^*\right)\left(A Y A^*\right)=P(M Y) A^*, \\ & \overline{P M A^* \times P N A^*}=\overline{{}^t \tilde{P}(M \times N){}^t \tilde{A}^*}}=P \overline{M} \times \overline{N} A^*, \text { etc. } \end{aligned}$
basis: inferred / ink					
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value					
Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
0902.0431_E00515 (was)	85	■0.002	$\begin{aligned} & \lambda_0 - \lambda_1 = 1 \quad \lambda_0 - \lambda_2 = 1 \quad \lambda_0 - \lambda_3 = 1 \\ & \lambda_0 - \lambda_1 = 1 \quad \lambda_0 - \lambda_2 = 1 \quad \lambda_0 - \lambda_3 = 1 \\ & \lambda_0 - \lambda_1 = 1 \quad \lambda_0 - \lambda_2 = 1 \quad \lambda_0 - \lambda_3 = 1 \end{aligned}$	$\begin{aligned} \lambda_0 - \lambda_1 &= 1 & 0 & 0 & 0 & 0 & 0 \\ \lambda_0 - \lambda_2 &= 1 & 1 & 0 & 0 & 0 & 0 \\ \lambda_0 - \lambda_3 &= 1 & 1 & 1 & 0 & 0 & 0 \end{aligned}$	$\begin{aligned} \lambda_0 + \lambda_1 &= 1 & 2 & 2 & 1 & 1 & 1 \\ \lambda_0 + \lambda_2 &= 1 & 1 & 2 & 1 & 1 & 1 \\ \lambda_0 + \lambda_3 &= 1 & 1 & 1 & 1 & 1 & 1 \end{aligned}$
0902.0431_E00515 (now)	85	—	$\begin{aligned} & \lambda_0 - \lambda_1 = 1 \quad \lambda_0 - \lambda_2 = 1 \quad \lambda_0 - \lambda_3 = 1 \\ & \lambda_0 - \lambda_1 = 1 \quad \lambda_0 - \lambda_2 = 1 \quad \lambda_0 - \lambda_3 = 1 \\ & \lambda_0 - \lambda_1 = 1 \quad \lambda_0 - \lambda_2 = 1 \quad \lambda_0 - \lambda_3 = 1 \end{aligned}$	$\begin{aligned} \lambda_0 - \lambda_1 &= 1 & 0 & 0 & 0 & 0 & 0 \\ \lambda_0 - \lambda_2 &= 1 & 1 & 0 & 0 & 0 & 0 \\ \lambda_0 - \lambda_3 &= 1 & 1 & 1 & 0 & 0 & 0 \end{aligned}$	$\begin{aligned} \lambda_0 + \lambda_1 &= 1 & 2 & 2 & 1 & 1 & 1 \\ \lambda_0 + \lambda_2 &= 1 & 1 & 2 & 1 & 1 & 1 \\ \lambda_0 + \lambda_3 &= 1 & 1 & 1 & 1 & 1 & 1 \end{aligned}$
basis: inferred / ink					
0902.0431_E00515 (was)	85	■0.002	$\begin{aligned} & \lambda_0 - \lambda_1 = 1 \quad \lambda_0 - \lambda_2 = 1 \quad \lambda_0 - \lambda_3 = 1 \\ & \lambda_0 - \lambda_1 = 1 \quad \lambda_0 - \lambda_2 = 1 \quad \lambda_0 - \lambda_3 = 1 \\ & \lambda_0 - \lambda_1 = 1 \quad \lambda_0 - \lambda_2 = 1 \quad \lambda_0 - \lambda_3 = 1 \end{aligned}$	$\begin{aligned} \lambda_0 - \lambda_1 &= 1 & 0 & 0 & 0 & 0 & 0 \\ \lambda_0 - \lambda_2 &= 1 & 1 & 0 & 0 & 0 & 0 \\ \lambda_0 - \lambda_3 &= 1 & 1 & 1 & 0 & 0 & 0 \end{aligned}$	$\begin{aligned} \lambda_0 + \lambda_1 &= 1 & 2 & 2 & 1 & 1 & 1 \\ \lambda_0 + \lambda_2 &= 1 & 1 & 2 & 1 & 1 & 1 \\ \lambda_0 + \lambda_3 &= 1 & 1 & 1 & 1 & 1 & 1 \end{aligned}$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
0902.0431_E00756 (now)	125	—	$\begin{array}{l} \backslash begin{array}{llllllllllllll} \lambda_{\{0\}}-\lambda_{\{1\}}=1 \\ \& 0 \& 0 \& 0 \& 0 \& 0 \& 0 \& 0 \& \lambda_{\{0\}}+\lambda_{\{1\}}=1 \& \\ 2 \& 2 \& 2 \& 2 \& 2 \& 1 \backslash \backslash \lambda_{\{0\}}-\lambda_{\{2\}}=1 \& 1 \& \\ 0 \& 0 \& 0 \& 0 \& 0 \& 0 \& \lambda_{\{0\}}+\lambda_{\{2\}}=1 \& 1 \& 2 \\ \& 2 \& 2 \& 1 \backslash \backslash \lambda_{\{0\}}-\lambda_{\{3\}}=1 \& 1 \& 1 \& 0 \\ \& 0 \& 0 \& 0 \& 0 \& \lambda_{\{0\}}+\lambda_{\{3\}}=1 \& 1 \& 1 \& 2 \& \\ 2 \& 1 \backslash \backslash \lambda_{\{1\}}-\lambda_{\{2\}}=0 \& 1 \& 0 \& 0 \& 0 \& 0 \& \\ 0 \& 0 \& \lambda_{\{1\}}+\lambda_{\{2\}}=0 \& 1 \& 2 \& 2 \& 2 \& 1 \\ \backslash \backslash \lambda_{\{1\}}-\lambda_{\{3\}}=0 \& 1 \& 1 \& 0 \& 0 \& 0 \& 0 \& 0 \\ \& \lambda_{\{1\}}+\lambda_{\{3\}}=0 \& 1 \& 1 \& 2 \& 2 \& 1 \backslash \backslash \\ \lambda_{\{2\}}-\lambda_{\{3\}}=0 \& 0 \& 1 \& 0 \& 0 \& 0 \& 0 \& 0 \& \\ \lambda_{\{2\}}+\lambda_{\{3\}}=0 \& 0 \& 1 \& 2 \& 2 \& 1 \backslash \backslash \& \& \\ \lambda_{\{0\}}+\frac{1}{2}(\mu_{\{2\}}-\mu_{\{3\}})\left(\mu_{\{2\}}-\mu_{\{3\}}\right)=1 \\ \& 1 \& 1 \& 1 \& 1 \& 1 \& 1 \& \& \& \& \backslash \backslash \& \& \lambda_{\{1\}}+ \\ \frac{1}{2}(\mu_{\{2\}}-\mu_{\{3\}})\left(\mu_{\{2\}}-\mu_{\{3\}}\right)=0 \& 1 \& 1 \& 1 \& 1 \& \\ 1 \& 1 \& 1 \& 1 \& \& \& \backslash \backslash \& \& \lambda_{\{2\}}+\frac{1}{2}(\mu_{\{2\}}-\mu_{\{3\}})\left(\mu_{\{2\}}-\mu_{\{3\}}\right)=0 \& 0 \& 1 \& 1 \& 1 \& 1 \& 1 \& \& \& \\ \& \backslash \backslash \& \& \lambda_{\{3\}}+\frac{1}{2}(\mu_{\{2\}}-\mu_{\{3\}})\left(\mu_{\{2\}}-\mu_{\{3\}}\right)=0 \& 0 \& 0 \& 1 \& 1 \& 1 \& 1 \& \& \& \& \end{array}$	$\begin{array}{llllll} \lambda_0-\lambda_1=1 & 0 & 0 & 0 & 0 & 0 \\ \lambda_0-\lambda_2=1 & 1 & 0 & 0 & 0 & 0 \\ \lambda_0-\lambda_3=1 & 1 & 1 & 0 & 0 & 0 \\ \lambda_1-\lambda_2=0 & 1 & 0 & 0 & 0 & 0 \\ \lambda_1-\lambda_3=0 & 1 & 1 & 0 & 0 & 0 \\ \lambda_2-\lambda_3=0 & 0 & 1 & 0 & 0 & 0 \end{array} \quad \begin{array}{llllll} 0 & 0 & 0 & 0 & \lambda_0+\lambda_1=1 & 2 \\ 0 & 0 & 0 & 0 & \lambda_0+\lambda_2=1 & 1 \\ 0 & 0 & 0 & 0 & \lambda_0+\lambda_3=1 & 1 \\ 0 & 0 & 0 & 0 & \lambda_1+\lambda_2=0 & 1 \\ 0 & 0 & 0 & 0 & \lambda_1+\lambda_3=0 & 1 \\ 0 & 0 & 0 & 0 & \lambda_2+\lambda_3=0 & 0 \end{array} \quad \begin{array}{llllll} 2 & 2 & 2 & 2 & 1 & 1 \\ 2 & 2 & 2 & 2 & 1 & 1 \\ 1 & 1 & 2 & 2 & 1 & 1 \\ 2 & 2 & 2 & 2 & 1 & 1 \\ 2 & 2 & 2 & 2 & 1 & 1 \\ 1 & 1 & 2 & 2 & 1 & 1 \end{array}$	$\begin{array}{llllll} \lambda_0+\lambda_1=1 & 2 & 2 & 2 & 2 & 1 \\ \lambda_0+\lambda_2=1 & 1 & 2 & 2 & 2 & 1 \\ \lambda_0+\lambda_3=1 & 1 & 1 & 2 & 2 & 1 \\ \lambda_1+\lambda_2=0 & 1 & 2 & 2 & 2 & 1 \\ \lambda_1+\lambda_3=0 & 1 & 1 & 2 & 2 & 1 \\ \lambda_2+\lambda_3=0 & 0 & 1 & 2 & 2 & 1 \end{array}$
basis: inferred / ink					
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
0902.0431_E00922 (was)	153	■0.102	f^{ -1}\left(\sum_{i<j<k}x_{i\ j\ k}\boldsymbol{e}_{_i}\wedge\boldsymbol{e}_{_j}\wedge\boldsymbol{e}_{_k}\right)=\left(\begin{array}{c}\left(\begin{array}{ccc}\left(\begin{array}{ccc}h(x_{156}) & h(x_{164},\overline{x}_{256}) & h(x_{145},\overline{x}_{356}) \\ h(x_{256},\overline{x}_{164}) & h(x_{264}) & h(x_{245},\overline{x}_{364}) \\ h(x_{356},\overline{x}_{145}) & h(x_{364},\overline{x}_{245}) & h(x_{345})\end{array}\right) \\ h(x_{423}) & h(x_{431},\overline{x}_{523}) & h(x_{412},\overline{x}_{623}) \\ h(x_{523},\overline{x}_{431}) & h(x_{531}) & h(x_{512},\overline{x}_{631}) \\ h(x_{623},\overline{x}_{412}) & h(x_{631},\overline{x}_{512}) & h(x_{612})\end{array}\right) \\ h(x_{123}) \\ h(x_{456})\end{array}\right),	(not rendered)	
0902.0431_E00922 (now)	153	—	f^{ -1}\left(\sum_{i<j<k}x_{ijk}\boldsymbol{e}_{_i}\wedge\boldsymbol{e}_{_j}\wedge\boldsymbol{e}_{_k}\right)=\left(\begin{array}{c}\left(\begin{array}{ccc}\left(\begin{array}{ccc}h(x_{156}) & h(x_{164},\overline{x}_{256}) & h(x_{145},\overline{x}_{356}) \\ h(x_{256},\overline{x}_{164}) & h(x_{264}) & h(x_{245},\overline{x}_{364}) \\ h(x_{356},\overline{x}_{145}) & h(x_{364},\overline{x}_{245}) & h(x_{345})\end{array}\right) \\ h(x_{423}) & h(x_{431},\overline{x}_{523}) & h(x_{412},\overline{x}_{623}) \\ h(x_{523},\overline{x}_{431}) & h(x_{531}) & h(x_{512},\overline{x}_{631}) \\ h(x_{623},\overline{x}_{412}) & h(x_{631},\overline{x}_{512}) & h(x_{612})\end{array}\right) \\ h(x_{123}) \\ h(x_{456})\end{array}\right),	(not rendered)	

basis: inferred / ink

Conf.

MathPix confidence:

■

≥0.9

■

0.5-0.9

■

<0.5

—

no value

Residual

render vs scan (inkdrill):

●

C component

●

W weak

●

S stable

●

N noise

●

K clean

●

not measured

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
0902.0431_E01032 (was)	173	■ 0.022	$\begin{aligned} & \frac{1}{2}\left(\lambda_0+\right. \\ & \left.\lambda_1+\lambda_2-\lambda_3\right)+\frac{1}{2} \\ & \mu_2-\frac{1}{3} \nu=1 \quad 2 \quad 3 \quad 3 \quad 2 \quad 2 \quad 1 \quad 1 \\ & \frac{1}{2}\left(\lambda_0+\lambda_1-\lambda_2+\lambda_3\right)+\frac{1}{2} \mu_2-\frac{2}{3} \nu=1 \quad 2 \quad 3 \quad 3 \quad 2 \quad 1 \quad 1 \quad 1 \\ & \frac{1}{2}\left(\lambda_0-\lambda_1+\lambda_2+\lambda_3\right)+\frac{1}{2} \mu_2-\frac{2}{3} \nu=1 \quad 2 \quad 3 \quad 3 \quad 2 \quad 1 \quad 0 \quad 1 \\ & \frac{1}{2}\left(\lambda_0-\lambda_1-\lambda_2-\lambda_3\right)+\frac{1}{2} \mu_2-\frac{2}{3} \nu=1 \quad 1 \quad 1 \quad 1 \quad 0 \quad 0 \quad 0 \quad 0 \\ & \frac{1}{2}\left(\lambda_0+\lambda_1+\lambda_2-\lambda_3\right)-\frac{1}{2} \mu_2+\frac{2}{3} \nu=1 \quad 2 \quad 3 \quad 2 \quad 2 \quad 2 \quad 1 \quad 2 \\ & \frac{1}{2}\left(\lambda_0+\lambda_1-\lambda_2+\lambda_3\right)-\frac{1}{2} \mu_2+\frac{2}{3} \nu=1 \quad 2 \quad 3 \quad 2 \quad 2 \quad 1 \quad 1 \quad 2 \\ & \frac{1}{2}\left(\lambda_0-\lambda_1+\lambda_2+\lambda_3\right)-\frac{1}{2} \mu_2-\frac{2}{3} \nu=1 \quad 2 \quad 3 \quad 2 \quad 2 \quad 1 \quad 0 \quad 2 \\ & \frac{1}{2}\left(\lambda_0-\lambda_1-\lambda_2-\lambda_3\right)-\frac{1}{2} \mu_2+\frac{2}{3} \nu=1 \quad 1 \quad 1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 1 \\ & \frac{1}{2}\left(\lambda_0+\lambda_1+\lambda_2+\lambda_3\right)+\frac{1}{2} \mu_3-\frac{2}{3} \nu=1 \quad 2 \quad 3 \quad 3 \quad 3 \quad 2 \quad 1 \quad 1 \\ & \frac{1}{2}\left(\lambda_0+\lambda_1-\lambda_2-\lambda_3\right)+\frac{1}{2} \mu_3-\frac{2}{3} \nu=1 \quad 1 \quad 1 \quad 1 \quad 1 \quad 1 \quad 1 \quad 0 \\ & \frac{1}{2}\left(\lambda_0-\lambda_1+\lambda_2-\lambda_3\right)+\frac{1}{2} \mu_3-\frac{2}{3} \nu=1 \quad 1 \quad 1 \quad 1 \quad 1 \quad 1 \quad 0 \quad 0 \\ & \frac{1}{2}\left(\lambda_0-\lambda_1-\lambda_2+\lambda_3\right)+\frac{1}{2} \mu_3-\frac{2}{3} \nu=1 \quad 1 \quad 1 \quad 1 \quad 1 \quad 0 \quad 0 \quad 0 \\ & \frac{1}{2}\left(\lambda_0+\lambda_1+\lambda_2+\lambda_3\right)-\frac{1}{2} \mu_3+\frac{2}{3} \nu=1 \quad 3 \quad 5 \quad 4 \quad 3 \quad 2 \quad 1 \quad 3 \\ & \frac{1}{2}\left(\lambda_0+\lambda_1-\lambda_2-\lambda_3\right)-\frac{1}{2} \mu_3+\frac{2}{3} \nu=1 \quad 2 \quad 3 \quad 2 \quad 1 \quad 1 \quad 1 \quad 2 \\ & \frac{1}{2}\left(\lambda_0-\lambda_1+\lambda_2-\lambda_3\right)-\frac{1}{2} \mu_3+\frac{2}{3} \nu=1 \quad 2 \quad 3 \quad 2 \quad 1 \quad 1 \quad 0 \quad 2 \\ & \frac{1}{2}\left(\lambda_0-\lambda_1-\lambda_2+\lambda_3\right)-\frac{1}{2} \mu_3+\frac{2}{3} \nu=1 \quad 2 \quad 3 \quad 2 \quad 1 \quad 0 \quad 0 \quad 2 \end{aligned}$	(not rendered)	
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
0902.0431_E01032 (now)	173	—	$\begin{aligned} \kappa\pi H &= \tfrac{1}{2}(-\lambda_0 + \lambda_1 - \lambda_2 + \lambda_3)H_0 + \tfrac{1}{2}(-\lambda_0 + \lambda_1 + \lambda_2 - \lambda_3)H_1 \\ &+ \tfrac{1}{2}(\lambda_0 + \lambda_1 + \lambda_2 + \lambda_3)H_2 + \tfrac{1}{2}(-\lambda_0 - \lambda_1 + \lambda_2 + \lambda_3)H_3, \end{aligned}$	$\kappa\pi H = \tfrac{1}{2}(-\lambda_0 + \lambda_1 - \lambda_2 + \lambda_3)H_0 + \tfrac{1}{2}(-\lambda_0 + \lambda_1 + \lambda_2 - \lambda_3)H_1 + \tfrac{1}{2}(\lambda_0 + \lambda_1 + \lambda_2 + \lambda_3)H_2 + \tfrac{1}{2}(-\lambda_0 - \lambda_1 + \lambda_2 + \lambda_3)H_3,$	$\begin{aligned} \frac{1}{2}(\lambda_0 + \lambda_1 + \lambda_2 - \lambda_3) + \frac{1}{2}\mu_2 - \frac{2}{3}\nu &= 1 & 2 & 3 & 3 & 2 & 2 & 1 & 1 \\ \frac{1}{2}(\lambda_0 + \lambda_1 - \lambda_2 + \lambda_3) + \frac{1}{2}\mu_2 - \frac{2}{3}\nu &= 1 & 2 & 3 & 3 & 2 & 1 & 1 & 1 \\ \frac{1}{2}(\lambda_0 - \lambda_1 + \lambda_2 + \lambda_3) + \frac{1}{2}\mu_2 - \frac{2}{3}\nu &= 1 & 2 & 3 & 3 & 2 & 1 & 0 & 1 \\ \frac{1}{2}(\lambda_0 - \lambda_1 - \lambda_2 - \lambda_3) + \frac{1}{2}\mu_2 - \frac{2}{3}\nu &= 1 & 1 & 1 & 1 & 0 & 0 & 0 & 0 \\ \frac{1}{2}(\lambda_0 + \lambda_1 + \lambda_2 - \lambda_3) - \frac{1}{2}\mu_2 + \frac{2}{3}\nu &= 1 & 2 & 3 & 2 & 2 & 2 & 1 & 2 \\ \frac{1}{2}(\lambda_0 + \lambda_1 - \lambda_2 + \lambda_3) - \frac{1}{2}\mu_2 + \frac{2}{3}\nu &= 1 & 2 & 3 & 2 & 2 & 1 & 1 & 2 \\ \frac{1}{2}(\lambda_0 - \lambda_1 + \lambda_2 + \lambda_3) - \frac{1}{2}\mu_2 - \frac{2}{3}\nu &= 1 & 2 & 3 & 2 & 2 & 1 & 0 & 2 \\ \frac{1}{2}(\lambda_0 - \lambda_1 - \lambda_2 - \lambda_3) - \frac{1}{2}\mu_2 + \frac{2}{3}\nu &= 1 & 1 & 1 & 0 & 0 & 0 & 0 & 1 \\ \frac{1}{2}(\lambda_0 + \lambda_1 + \lambda_2 + \lambda_3) + \frac{1}{2}\mu_3 - \frac{2}{3}\nu &= 1 & 2 & 3 & 3 & 3 & 2 & 1 & 1 \\ \frac{1}{2}(\lambda_0 + \lambda_1 - \lambda_2 - \lambda_3) + \frac{1}{2}\mu_3 - \frac{2}{3}\nu &= 1 & 1 & 1 & 1 & 1 & 1 & 1 & 0 \\ \frac{1}{2}(\lambda_0 - \lambda_1 + \lambda_2 - \lambda_3) + \frac{1}{2}\mu_3 - \frac{2}{3}\nu &= 1 & 1 & 1 & 1 & 1 & 1 & 0 & 0 \\ \frac{1}{2}(\lambda_0 - \lambda_1 - \lambda_2 + \lambda_3) + \frac{1}{2}\mu_3 - \frac{2}{3}\nu &= 1 & 1 & 1 & 1 & 1 & 0 & 0 & 0 \\ \frac{1}{2}(\lambda_0 + \lambda_1 + \lambda_2 + \lambda_3) - \frac{1}{2}\mu_3 + \frac{2}{3}\nu &= 1 & 3 & 5 & 4 & 3 & 2 & 1 & 3 \\ \frac{1}{2}(\lambda_0 + \lambda_1 - \lambda_2 - \lambda_3) - \frac{1}{2}\mu_3 + \frac{2}{3}\nu &= 1 & 2 & 3 & 2 & 1 & 1 & 1 & 2 \\ \frac{1}{2}(\lambda_0 - \lambda_1 + \lambda_2 - \lambda_3) - \frac{1}{2}\mu_3 + \frac{2}{3}\nu &= 1 & 2 & 3 & 2 & 1 & 1 & 0 & 2 \\ \frac{1}{2}(\lambda_0 - \lambda_1 - \lambda_2 + \lambda_3) - \frac{1}{2}\mu_3 + \frac{2}{3}\nu &= 1 & 2 & 3 & 2 & 1 & 0 & 0 & 2 \end{aligned}$
basis: inferred / ink					
0902.0431_E01059 (was)	180	■ 0.093	$\begin{aligned} &\&=\left\{\alpha\in\left(E_8\right)^{\left\{1,1-,1-\right\}}\mid\tau\widetilde{\lambda}\alpha=\alpha\tau\widetilde{\lambda}\right\} \\ &\&=\left\{\alpha\in E_7^C\mid\tau\widetilde{\lambda}\alpha=\alpha\tau\widetilde{\lambda}\right\}\text{ (Proposition 5.7.1)} \\ &\&=\left\{\alpha\in E_7^C\mid\tau\lambda\alpha=\alpha\tau\lambda\right\}\text{ (by the correspondence to Proposition 5.7.1)} \\ &\&=E_7\text{ (Lemma 4.3.3.(4)).} \end{aligned}$	$\begin{aligned} &=\left\{\alpha\in\left(E_8^C\right)_{1,1-,1-}\mid\tau\widetilde{\lambda}\alpha=\alpha\tau\widetilde{\lambda}\right\} \\ &=\left\{\alpha\in E_7^C\mid\tau\widetilde{\lambda}\alpha=\alpha\tau\widetilde{\lambda}\right\}\text{ (Proposition 5.7.1)} \\ &=\left\{\alpha\in E_7^C\mid\tau\lambda\alpha=\alpha\tau\lambda\right\}\text{ (by the correspondence to Proposition 5.7.1)} \\ &=E_7\text{ (Lemma 4.3.3.(4)).} \end{aligned}$	$\begin{aligned} &=\left\{\alpha\in\left(E_8^C\right)_{1,1-,1-}\mid\tau\widetilde{\lambda}\alpha=\alpha\tau\widetilde{\lambda}\right\} \\ &=\left\{\alpha\in E_7^C\mid\tau\widetilde{\lambda}\alpha=\alpha\tau\widetilde{\lambda}\right\}\text{ (Proposition 5.7.1)} \\ &=\left\{\alpha\in E_7^C\mid\tau\lambda\alpha=\alpha\tau\lambda\right\}\text{ (by the correspondence to Proposition 5.7.1)} \\ &=E_7\text{ (Lemma 4.3.3.(4)).} \end{aligned}$
Conf. MathPix confidence: ■ ≥ 0.9 ■ 0.5–0.9 ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
0902.0431_E01059 (now)	180	—	$\begin{aligned} \kappa\pi H = & \frac{1}{2}(-\lambda_0 + \lambda_1 - \lambda_2 + \lambda_3)H_0 + \frac{1}{2}(-\lambda_0 + \lambda_1 + \lambda_2 - \lambda_3)H_1 \\ & + \frac{1}{2}(\lambda_0 + \lambda_1 + \lambda_2 + \lambda_3)H_2 + \frac{1}{2}(-\lambda_0 - \lambda_1 + \lambda_2 + \lambda_3)H_3, \end{aligned}$	$\begin{aligned} &= \{\alpha \in (E_8^{\,C})_{1,1^-,1_-} \mid \tau\widetilde{\lambda}\alpha = \alpha\tau\widetilde{\lambda}\} \\ &= \{\alpha \in E_7^{\,C} \mid \tau\widetilde{\lambda}\alpha = \alpha\tau\widetilde{\lambda}\} \text{ (Proposition 5.7.1)} \\ &= \{\alpha \in E_7^{\,C} \mid \tau\lambda\alpha = \alpha\tau\lambda\} \text{ (by the correspondence to Proposition 5.7.1)} \\ &= E_7 \text{ (Lemma 4.3.3.(4)).} \end{aligned}$	
basis: inferred / ink					
0902.0431_E01117 (was)	189	■ 0.108	$\begin{aligned} &\& (\boldsymbol{X}, \boldsymbol{Y}) = \\ &\left(X_{\{1\}}, Y_{\{1\}} \right) + \left(X_{\{2\}}, Y_{\{2\}} \right) + \\ &\left(X_{\{3\}}, Y_{\{3\}} \right) \in C, \quad \& \& \angle \\ &\boldsymbol{X}, \boldsymbol{Y} \angle = \& \angle \\ &X_{\{1\}}, Y_{\{1\}} \angle + \& \angle X_{\{2\}}, Y_{\{2\}} \\ &\angle + \& \angle X_{\{3\}}, Y_{\{3\}} \angle \\ &\in C, \quad \& \boldsymbol{X} \times \boldsymbol{Y} = \begin{pmatrix} X_2 \times Y_3 - Y_2 \times X_3 \\ X_3 \times Y_1 - Y_3 \times X_1 \\ X_1 \times Y_2 - Y_1 \times X_2 \end{pmatrix} \in (\mathfrak{J}^C)^3, \\ &\boldsymbol{X} \cdot \boldsymbol{Y} = \begin{pmatrix} (X_1, Y_1) & (X_1, Y_2) & (X_1, Y_3) \\ (X_2, Y_1) & (X_2, Y_2) & (X_2, Y_3) \\ (X_3, Y_1) & (X_3, Y_2) & (X_3, Y_3) \end{pmatrix} - \frac{1}{3}(\boldsymbol{X}, \boldsymbol{Y})E \in \mathfrak{sl}(3, C), \\ &\boldsymbol{X} \vee \boldsymbol{Y} = X_1 \vee Y_1 + X_2 \vee Y_2 + X_3 \vee Y_3 \in \mathfrak{e}_6^{\,C}, \end{aligned}$	$\begin{aligned} &(\boldsymbol{X}, \boldsymbol{Y}) = (X_1, Y_1) + (X_2, Y_2) + (X_3, Y_3) \in C, \\ &\langle \boldsymbol{X}, \boldsymbol{Y} \rangle = \langle X_1, Y_1 \rangle + \langle X_2, Y_2 \rangle + \langle X_3, Y_3 \rangle \in C, \\ &\boldsymbol{X} \times \boldsymbol{Y} = \begin{pmatrix} X_2 \times Y_3 - Y_2 \times X_3 \\ X_3 \times Y_1 - Y_3 \times X_1 \\ X_1 \times Y_2 - Y_1 \times X_2 \end{pmatrix} \in (\mathfrak{J}^C)^3, \\ &\boldsymbol{X} \cdot \boldsymbol{Y} = \begin{pmatrix} (X_1, Y_1) & (X_1, Y_2) & (X_1, Y_3) \\ (X_2, Y_1) & (X_2, Y_2) & (X_2, Y_3) \\ (X_3, Y_1) & (X_3, Y_2) & (X_3, Y_3) \end{pmatrix} - \frac{1}{3}(\boldsymbol{X}, \boldsymbol{Y})E \in \mathfrak{sl}(3, C), \\ &\boldsymbol{X}Y_1 + X_2 \vee Y_2 + X_3 \vee Y_3 \in \mathfrak{e}_6^{\,C}, \end{aligned}$	$\begin{aligned} &(\boldsymbol{X}, \boldsymbol{Y}) = (X_1, Y_1) + (X_2, Y_2) + (X_3, Y_3) \in C, \\ &\langle \boldsymbol{X}, \boldsymbol{Y} \rangle = \langle X_1, Y_1 \rangle + \langle X_2, Y_2 \rangle + \langle X_3, Y_3 \rangle \in C, \\ &\boldsymbol{X} \times \boldsymbol{Y} = \begin{pmatrix} X_2 \times Y_3 - Y_2 \times X_3 \\ X_3 \times Y_1 - Y_3 \times X_1 \\ X_1 \times Y_2 - Y_1 \times X_2 \end{pmatrix} \in (\mathfrak{J}^C)^3, \\ &\boldsymbol{X} \cdot \boldsymbol{Y} = \begin{pmatrix} (X_1, Y_1) & (X_1, Y_2) & (X_1, Y_3) \\ (X_2, Y_1) & (X_2, Y_2) & (X_2, Y_3) \\ (X_3, Y_1) & (X_3, Y_2) & (X_3, Y_3) \end{pmatrix} - \frac{1}{3}(\boldsymbol{X}, \boldsymbol{Y})E \in \mathfrak{sl}(3, C), \\ &\boldsymbol{X} \vee \boldsymbol{Y} = X_1 \vee Y_1 + X_2 \vee Y_2 + X_3 \vee Y_3 \in \mathfrak{e}_6^{\,C}, \end{aligned}$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
0902.0431_E01117 (now)	189	—	$\begin{aligned} & (\boldsymbol{X}, \boldsymbol{Y}) = (X_1, Y_1) + (X_2, Y_2) + (X_3, Y_3) \in C, \\ & \langle \boldsymbol{X}, \boldsymbol{Y} \rangle = \langle X_1, Y_1 \rangle + \langle X_2, Y_2 \rangle + \langle X_3, Y_3 \rangle \in C, \\ & \boldsymbol{X} \times \boldsymbol{Y} = \begin{pmatrix} X_2 \times Y_3 - Y_2 \times X_3 \\ X_3 \times Y_1 - Y_3 \times X_1 \\ X_1 \times Y_2 - Y_1 \times X_2 \end{pmatrix} \in (\mathfrak{J}^C)^3, \\ & \boldsymbol{X} \wedge \boldsymbol{Y} = \begin{pmatrix} X_2 \wedge Y_3 - Y_2 \wedge X_3 \\ X_3 \wedge Y_1 - Y_3 \wedge X_1 \\ X_1 \wedge Y_2 - Y_1 \wedge X_2 \end{pmatrix} \in (\mathfrak{J}^C)^3, \\ & \boldsymbol{X} \cdot \boldsymbol{Y} = \begin{pmatrix} (X_1, Y_1) & (X_1, Y_2) & (X_1, Y_3) \\ (X_2, Y_1) & (X_2, Y_2) & (X_2, Y_3) \\ (X_3, Y_1) & (X_3, Y_2) & (X_3, Y_3) \end{pmatrix} - \frac{1}{3}(\boldsymbol{X}, \boldsymbol{Y})E \in \mathfrak{sl}(3, C), \\ & \boldsymbol{X} \vee \boldsymbol{Y} = X_1 \vee Y_1 + X_2 \vee Y_2 + X_3 \vee Y_3 \in \mathfrak{e}_6^C, \end{aligned}$	$\begin{aligned} & (\boldsymbol{X}, \boldsymbol{Y}) = (X_1, Y_1) + (X_2, Y_2) + (X_3, Y_3) \in C, \\ & \langle \boldsymbol{X}, \boldsymbol{Y} \rangle = \langle X_1, Y_1 \rangle + \langle X_2, Y_2 \rangle + \langle X_3, Y_3 \rangle \in C, \\ & \boldsymbol{X} \times \boldsymbol{Y} = \begin{pmatrix} X_2 \times Y_3 - Y_2 \times X_3 \\ X_3 \times Y_1 - Y_3 \times X_1 \\ X_1 \times Y_2 - Y_1 \times X_2 \end{pmatrix} \in (\mathfrak{J}^C)^3, \\ & \boldsymbol{X} \wedge \boldsymbol{Y} = \begin{pmatrix} X_2 \wedge Y_3 - Y_2 \wedge X_3 \\ X_3 \wedge Y_1 - Y_3 \wedge X_1 \\ X_1 \wedge Y_2 - Y_1 \wedge X_2 \end{pmatrix} \in (\mathfrak{J}^C)^3, \\ & \boldsymbol{X} \cdot \boldsymbol{Y} = \begin{pmatrix} (X_1, Y_1) & (X_1, Y_2) & (X_1, Y_3) \\ (X_2, Y_1) & (X_2, Y_2) & (X_2, Y_3) \\ (X_3, Y_1) & (X_3, Y_2) & (X_3, Y_3) \end{pmatrix} - \frac{1}{3}(\boldsymbol{X}, \boldsymbol{Y})E \in \mathfrak{sl}(3, C), \\ & \boldsymbol{X} \vee \boldsymbol{Y} = X_1 \vee Y_1 + X_2 \vee Y_2 + X_3 \vee Y_3 \in \mathfrak{e}_6^C, \end{aligned}$	
basis: inferred / ink					
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					